

Shaun Allison

shallison@protonmail.com • <https://shalliso.github.io/> • <https://github.com/shalliso>
Toronto, Canada
US citizen and Canadian permanent resident

Mathematical logician and software engineer interested in AI-assisted mathematics, formal verification, and machine learning. Five papers in descriptive set theory, including the *Journal of the European Mathematical Society*, *Advances in Mathematics*, and the *Journal of Mathematical Logic*; Lean 4 formalizations of mathematics; and 14 months as a software engineer at Meta, with two granted patents on machine-learned ranking.

Tools: Lean 4 with Mathlib, Python, $\text{\LaTeX}$, git, Pytorch; familiar with numpy, pandas, matplotlib; earlier industry work in Hack, JavaScript, Haskell, and C.

Formal Verification

Knight model formalization in Lean 4 and Mathlib, 2026

<https://github.com/shalliso/KnightModel>

Julia Knight's original result was an ad-hoc construction of a structure characterizing $\aleph_1$. I generalized this and formalized the resulting theory in Lean.

First stage of a project to build higher analogues characterizing $\aleph_\alpha$ for every $\alpha < \omega_1$ with AI assistance.

Approximately 5,700 lines of Lean, developed with substantial AI assistance (Claude) under my direction; the axiomatization, proof architecture, and mathematical review are my own.

Presented at the Fields Institute Set Theory Seminar (March 2026), the Cornell Logic Seminar (March 2026), and the University of Vienna Logic Colloquium (April 2026).

Automatic continuity in Lean 4 and Mathlib, 2025

https://github.com/shalliso/automatic_continuity

Complete formalization of an automatic continuity theorem: every Borel-measurable homomorphism from a Polish group into a separable topological group is continuous.

Approximately 300 lines of Lean, written by hand.

Employment

Independent Research, January 2026 – present

- Built the Lean 4 formalization described above
- Completed revisions on two papers accepted in 2026 (JEMS, JML)
- Research visits to Cornell University and the University of Vienna; upcoming visit to the Institute for Advanced Study in Mathematics, Zhejiang University
- Writing a mathematical logic textbook with Spencer Unger

Sessional Lecturer I, University of Toronto, September 2025 – December 2025

Postdoctoral Fellow, University of Toronto, August 2022 – July 2025

Postdoctoral Fellow, Hebrew University of Jerusalem, July 2021 – June 2022

Software Engineer, Meta (then Facebook), June 2015 – August 2016

- Games team: machine-learned targeting of ads and notifications — deciding which users to reach and with what. Two patents granted.

Software Engineering Intern, Meta, Summer 2014

Research and Development Intern, VMware, Summer 2013

Intern, Cisco, Summer 2012

Patents

Automatic recipient targeting for notifications (US10506055B2), with L. Foged. Granted. Machine-learned selection of which users should receive a given notification.

Selecting assets (US11019177B2), with L. Foged. Granted. Machine-learned selection of which creative asset to serve.

Publications and Preprints

1. *The class and dynamics of α -balanced Polish groups*, with Aristotelis Panagiotopoulos.
Journal of the European Mathematical Society, 2026. <https://doi.org/10.4171/JEMS/1798>
2. *Countable Borel treeable equivalence relations are classifiable by ℓ_1* .
Advances in Mathematics 479, 2025. <https://doi.org/10.1016/j.aim.2025.110452>
3. *Classification strength of Polish groups: Involving S_∞* .
Journal of Mathematical Logic, 2026. <https://doi.org/10.1142/S0219061326500054>
4. *Actions of tame abelian product groups*, with Assaf Shani.
Journal of Mathematical Logic 23(3), 2023. <https://doi.org/10.1142/S0219061322500283>
5. *Dynamical obstructions to classification by (co)homology and other TSI-group invariants*, with Aristotelis Panagiotopoulos.
Transactions of the American Mathematical Society 374, 2021. <https://doi.org/10.1090/tran/8475>
6. *Non-Archimedean TSI Polish groups and their potential Borel complexity spectrum*.
Preprint, 2020. <https://arxiv.org/abs/2010.05085>

In preparation: a textbook on mathematical logic, with Spencer Unger.

Education

Ph.D. Mathematics, Carnegie Mellon University, 2021

– Dissertation: *Classification of equivalence relations by Polish group actions*. Advisor: Clinton Conley

B.S. Computer Science, Carnegie Mellon University, 2015

– Thesis: *A secure human-computable authentication scheme from the k -Junta problem*. Advisors: Manuel Blum and Jeremiah Blocki

B.S. Mathematics (Discrete Mathematics and Logic), Carnegie Mellon University, 2015

Awards

Putnam Competition, top 500 (2014)

Carnegie Mellon University Honors; College Honors (Computer Science)

Phi Beta Kappa

Selected Talks

Talks on the Lean formalization are listed under Formal Verification above. Full list available on request.

Polish automorphism groups that don't involve S_∞ , University of Vienna Set Theory Seminar (May 2026)

The class of CLI Polish groups is coanalytic-complete, North American Descriptive Set Theory Meeting, Brin Mathematics Research Center (October 2025)

A hierarchy of CLI Polish groups and non-reducibility results, Caltech Logic Seminar (November 2024)

A rank determining when a Polish automorphism group has less-than-maximal classification strength, European Set Theory Conference (September 2024)

Polish groups involving S_∞ , Structures: Descriptive Set Theory and Dynamics, Institute of Mathematics of the Polish Academy of Sciences (August 2023)

Polish groups which don't involve S_∞ , North American Annual Meeting of the Association for Symbolic Logic, UC Irvine (March 2023)

Polish groups with the pinned property, CMS Winter Meeting, Canadian Mathematical Society (December 2020)

Applications of the Gandy-Harrington topology, Boise Extravaganza in Set Theory, AAAS Pacific Division (June 2019)

Teaching

Instructor of record for eight courses at the University of Toronto and Carnegie Mellon University (2018–2025), from Differential Calculus (approx. 400 students) to upper-year Topology, Logic, and Field theory.

Supervised an undergraduate independent study in model theory (Winter 2023)

Founded a weekly graduate seminar in set theory and model theory at Carnegie Mellon University (2019)